MILITARY SPECIFICATION
CHAFF, COUNTERMEASURES, AIR LAUNCH, TYPE RR-( )/AL
GENERAL SPECIFICATION FOR

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes requirements for the design, manufacture, performance and acceptance of the Chaff, Countermeasures, Air Launch, Type RR-( )/AL, herein referred to as the Air Launch Chaff Countermeasures. The Air Launch Chaff Countermeasures (radar reflector) contains components capable of creating confusing returns to radar systems operating within specified frequency ranges.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering Specifications and Standardization Department (SESD) Code 53, Lakehurst, N.J. 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATION

Military

MIL-P-116	Preservation Packaging, Method of
DOD-D-1000	Drawings, Engineering and Associated List
MIL-T-18303	Test Procedures; Preproduction, Acceptance and Life for Aircraft Electronic Equipment, Format for
MIL-C-81745	"A" Size Sonobuoy Launch Container (SLC) LAU-126/A
MIL-C-85699/1	Chaff, Countermeasures, Air Launch (Type RR-181/AL)

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
DOD-STD-481	Configuration Control-Engineering Changes, Deviation and Waivers (Short Form)
MIL-STD-794	Parts and Equipment, Procedures for Packaging of
MIL-STD-810	Environmental Test Methods
MIL-STD-1323	Fleet Issue Unit Load
MIL-STD-2071	Testing of Chaff Radar Cross-Section

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein.

MANUALS

Naval Air Systems Command

NAVAIR 11-120-63	Ship Weapon Installation Manual, AN/ALQ 190(V) Countermeasures Set, Air Launch Countermeasures System

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PUBLICATIONS

Code of Federal Regulations

CFR 49 Parts 171-173 Transportation

(Copies of specifications, standards, other Government documents, drawing(s) and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

STANDARDS

American Society of Mechanical Engineers

ANSI Y14.5 Dimensioning and Tolerancing for Engineering Drawings

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, N.Y. 10017.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specification, specification sheet or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the applicable specification sheet, the latter shall govern.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2, 6.2.1 and 6.3).

3.3 Chaff material. The chaff material shall be in accordance with the applicable specification sheet (see 3.1).

3.3.1 Dipoles. Dipoles (see 6.4.6) shall meet the dimension and composition requirements of the applicable specification sheets (see 3.1). Dipoles composed of coated filaments shall have a d.c. resistivity not greater than 10 ohms per linear inch.

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3.3.2 Secondary (slip) coating. When specified in the applicable specification sheet (see 3.1), a secondary (slip) coating (see 6.4.15) shall be applied to the chaff material to:

a. Prevent dipole end welding and metal smear (see 6.4.11) during the cutting operation.

b. Prevent cohesion between dipoles which may be caused by corrosive conditions.

c. Increase dispersion by reducing dipole-to-dipole friction.

The formulation of the slip coating shall be approved by the procuring activity prior to use as part of the first article inspection procedures or quality conformance inspection procedures if a first article is not required.

3.4 Design and construction. Each Air Launch Chaff Countermeasures (see 6.4.1) including chaff payload (see 6.4.3) and all parts, assemblies, markings, packaging and packing; shall conform to all the requirements of this specification, and the applicable DL for design, construction and workmanship, except as otherwise specified in the contract (see 6.2.1). Configuration control shall be maintained in accordance with DOD-STD-481.

3.4.1 Nomenclature and identification markings. Nomenclature and identification markings shall be in accordance with MIL-STD-129, MIL-STD-130, CFR 49 Parts 171-173 and as specified herein.

3.4.2 Item description. The Air Launch Chaff Countermeasures (Type RR-( )/AL) is comprised of a canister, payload release assembly, safe and arm assembly and a cylindrical, frangible case containing the chaff payload. The chaff payload is comprised of aluminum coated glass dipoles. The dipoles are cut (see 6.4.4) to respond to specified radar threat frequencies. At a point after launch from the aircraft, dependent on relative wind velocity, the release assembly arms the safe and arm assembly and releases the chaff payload from the canister. When the chaff payload drops clear of the canister the safe and arm assembly detonates the center burster. The chaff payload is ejected through the frangible case by the center burster and the chaff dipole muffins (see 6.4.12) are dispersed. The dipoles produce a chaff cloud, creating a simulated target or decoy.

3.4.3 Physical characteristics. The Air Launch Chaff Countermeasures shall be functionally compatible with the Chaff Launch Container (see 6.4.2) (similar to the "A" size Sonobuoy Launch Container (SLC)(LAU-126/A) in accordance with MIL-C-81745) and the "A" size Sonobuoy Gravity Launch Dispenser when used with the aircraft as specified in the applicable specification sheet (see 3.1). Neither the operational reliability of the launcher or the effectiveness of the chaff payload shall be degraded when the Air Launch Chaff Countermeasures is used with these launchers or aircraft.

3.4.4 Launch envelope. Launching shall be possible utilizing any combination of altitude and airspeed within the launch envelope as specified in the applicable specification sheet (see 3.1).

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3.5 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions:

a. Temperature: Room ambient (25 degrees +/- 10 degrees C)

b. Altitude : Normal ground

c. Vibration : None

d. Humidity : Room ambient up to 90 percent relative humidity

3.6 Service conditions. The Air Launch Chaff Countermeasures shall operate as required under any of the environmental service conditions or combinations of conditions specified in MIL-STD-810 as modified herein. Tolerances shall be in accordance with the general requirements of MIL-STD-810.

3.7 Performance requirements. Unless otherwise specified herein, values set forth to establish the specified performance apply to performance under both standard and extreme service conditions. When reduced performance under extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance and standard conditions will be specified in the contract.

3.7.1 Performance characteristics. Performance characteristics shall be in accordance with MIL-STD-2071 and this specification when the Air Launch Chaff Countermeasures is tested in accordance with the requirements specified herein.

3.8 Detail requirements. The Air Launch Chaff Countermeasures contents shall meet the design, construction and physical requirements specified herein. Drawings shall be interpreted with DOD-D-1000 and ANSI Y14.5.

3.8.1 Components. Each Air Launch Chaff Countermeasures shall consist of a windflap, canister, payload release assembly, chaff payload assembly, and other components and shall be of the material, dimensions, quality and quantity specified in this specification and the applicable DL.

3.8.1.1 Windflap. The windflap shall initiate the payload release assembly (see 3.8.1.3) after the Air Launch Chaff Countermeasures is launched from the aircraft.

3.8.1.2 Canister. The canister shall consist of a cylindrical metal tube to which the payload release assembly (see 3.8.1.3) and chaff payload assembly (see 3.8.1.4) can be attached.

3.8.1.3 Payload release assembly. The payload release assembly shall consist of a device(s) as specified in the applicable specification sheet (see 3.1) capable of releasing and arming the the chaff payload assembly (see 3.8.1.4).

3.8.1.4 Chaff payload assembly. The chaff payload assembly shall consist of a safe and arm assembly, frangible case, center burster, and chaff payload.

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3.8.1.4.1 Safe and arm assembly. The safe and arm assembly shall be capable of being armed by the release assembly and detonating the center burster when the payload assembly exits the canister.

3.8.1.4.2 Frangible case. The frangible case shall consist of a material that fragments when the center burster charge is detonated.

3.8.1.4.3 Center burster. The center burster shall detonate and propel the chaff payload through the frangible case.

3.8.1.4.4 Chaff payload. The chaff payload shall be as specified in the applicable specification sheet (see 3.1).

3.8.2 Markings. The Air Launch Chaff Countermeasures shall be marked in accordance with the requirements of MIL-STD-130, CFR 49, Parts 171-173 and the applicable DL.

3.8.3 Assembly(s). The Air Launch Chaff Countermeasures shall be assembled in accordance with the requirements of this specification and the applicable DL. The chaff muffins and muffin packs (see 6.4.13) shall be positioned within the chaff assembly(s) as specified in the applicable DL.

3.8.4 Weight. The minimum gross weight of the Air Launch Chaff Countermeasures, chaff payload and chaff assembly(s) shall be as specified in the applicable specification sheet (see 3.1).

3.9 Workmanship. The Air Launch Chaff Countermeasures, including all parts shall be manufactured in accordance with the applicable DL and as specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. The Government will conduct all flight tests.

4.1.1 Responsibility for compliance. All items must meet all requirements of Section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality assurance program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Classification of inspections. The inspections requirements specified herein are classified as follows:

a. First Article Inspection (see 4.2).

b. Initial Production Inspection (see 4.3).

c. Quality Conformance Inspection (see 4.4).

4.1.3 Lot information. A lot shall consist of a group of items of a specific configuration, produced within the same time frame at one location, by the same production line and production technique, and submitted for acceptance, inspection, or use at one time.

4.2 First article inspection. First article inspection shall be conducted by the contractor on the Air Launch Chaff Countermeasures, chaff payload or chaff assembly(s) that are representative of the production units to be supplied under the contract. The Air Launch Chaff Countermeasures, chaff payload or chaff assembly(s) shall be inspected according to the approved first article inspection procedure (see 4.5). No first article inspection shall be conducted prior to acceptance of the first article test procedure by the procuring activity. First article samples shall not be submitted for functional (Government) tests until in-house tests have been conducted in accordance with the approved test procedures and the test results, prepared in accordance with MIL-STD-831 have been accepted by the procuring activity.

4.2.1 Scope of inspection. Twelve Air Launch Chaff Countermeasures shall be selected for first article inspection (see Table I) to verify that the Air Launch Chaff Countermeasures meets all the requirements of this specification, applicable DL and the contract (see 6.2.1). The tests on each group shall be run in the order specified in TABLE I. Each of the groups shall be clearly marked on each item before starting first article tests.

4.2.2 First article approval. Approval of the first article samples will be by the procuring activity upon satisfactory completion of all inspections. No production Air Launch Chaff Countermeasures shall be delivered prior to the approval of the first article samples.

4.2.3 Production air launch chaff countermeasures. Air Launch Chaff Countermeasures supplied under the contract shall be in all respects, including design, construction, workmanship, performance and quality be equal to the approved first article sample. Each Air Launch Chaff Countermeasures shall be capable of successfully passing the same tests as imposed on the first article samples. Evidence of noncompliance with the above shall constitute cause for rejection, and it shall be the obligation of the contractor to make the necessary corrections as approved by the procuring activity.

4.3 Initial production inspections. Two, four if first article is waived, of the first 50 production Air Launch Chaff Countermeasures shall be selected and sent at the contractor's expense to the designated Government Laboratory for inspection. No other inspections other than those listed in Table I and in accordance with TABLE II shall be conducted on the samples prior to starting the initial production inspections. The first article units shall

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not be selected for this inspection. Inspection procedures shall include, but not limited to, the procedures of 4.6 thru 4.10.

4.3.1 Scope of inspections. The samples shall be subjected to any and all inspections the procuring activity deems necessary to assure that the production units are equal to the previously approved first article samples in design, construction, workmanship, performance and quality and that they meet all requirements specified herein and in the contract.

4.3.2 Accessory material. In addition to the complete samples submitted for initial production inspections, the contractor shall also submit such accessory material and include data necessary to evaluate the samples in accordance with this specification and the contract (see 6.2.1).

TABLE I. First article inspections-schedule and sequence.

Inspection	Reqm't para.	AIR LAUNCH CHAFF CM Groups/(Qty) Insp. para.	1 (4)	2 (2)	3 (2)	4 (1)	5 (1)	6 (1)	7 (1)
Chaff material	3.3	4.6	X	X	X	X	X	X	X
Visual insp.	3.4	4.7	X	X	X	X	X	X	X
Assembly insp.	3.8.3	4.8.1	X	X	X	X	X	X	X
Weight	3.8.4	4.8.2	X	X	X	X	X	X	X
Temperature and altitude	3.6	4.9.1			X				
Humidity	3.6	4.9.2				X			
Salt fog	3.6	4.9.3					X		
Shock	3.6	4.9.4						X	
Vibration	3.6	4.9.5							X
Functional	3.7	4.10	X		X	X	X	X	X

4.3.3 Initial production sample approval. Approval of the initial production sample will be by the procuring activity upon satisfactory completion of all inspections. Any design, material, or performance defect made evident during these inspections shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the inspections shall be cause for deliveries under the contract to cease until proper corrective action is approved and accomplished.

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TABLE II. Initial production inspections-schedule and sequence.

Inspection	Reqm't Para.	AIR LAUNCH CHAFF CM Group/(Qty) (See Note 1) Insp. Para.	IA (2)	IB (2)
Chaff material	3.3	4.6	X	X
Visual insp.	3.4	4.7	X	X
Assembly insp.	3.8.3	4.8.1	X	X
Weight	3.8.4	4.8.2	X	X
Functional test	3.7	4.10		X

Note 1: Group IB is required only if first article has been waived.

4.4 Quality conformance inspections. The contractor shall furnish all samples and with the exception of Government performed flight tests, shall be responsible for accomplishing the quality conformance inspections. All examinations and tests shall be under the supervision of the Government representative. Acceptance or approval of material during the course of manufacturing shall not be construed as a guarantee of the acceptance of the finished product. Quality conformance inspections shall consist of the following:

a. Individual Inspections (see 4.4.1),

b. Sampling Inspections (see 4.4.2) and

c. Special Inspections (see 4.4.3).

4.4.1 Individual inspections. Individual inspections are those inspections conducted on each Air Launch Chaff Countermeasures. Individual inspections shall be as specified in TABLE lII. Failure to pass any of these inspections shall be cause for rejection of the Air Launch Chaff Countermeasures.

4.4.2 Sampling inspections. Air Launch Chaff Countermeasures selected for sampling inspections shall first have passed the individual inspections. The Air Launch Chaff Countermeasures shall be selected for sampling inspections by the Government representative in accordance with the schedule specified herein. The Government representative shall be informed as to when and where the test(s) will be conducted.

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4.4.2.1 Scope of inspections. Sampling inspections shall include examinations and tests for verification of manufacturing and performance requirements.

4.4.2.1.1 Sampling inspection for verification of manufacturing processes. Sampling inspections for verification of the manufacturing process shall include, as a minimum, those in-process inspections specified in 4.6, 4.7 and 4.8.

TABLE III. Individual inspection.

Inspection	Requirements Paragraph	Inspection Paragraph
Components	3.8.1	4.7.2
Markings	3.8.2	4.7.3
Assembly insp.	3.8.3	4.8.1
Weight	3.8.4	4.8.2
Packaging and packing	5.2, 5.3	4.7.4

4.4.2.1.2 Sampling inspection for verification of performance requirements.

4.4.2.1.2.1 Sampling inspections for verification of performance requirements (Government performed flight test). Sampling inspections for the verification of performance requirements (Flight Test) shall be conducted on Air Launch Chaff Countermeasures to demonstrate conformance to the requirements of 4.10 when conducted in accordance with the procedures of MIL-STD-2071. The contractor shall provide five Air Launch Chaff Countermeasures from each delivery lot (see 6.4.7).

4.4.3 Special inspections. Special inspections shall be conducted on five Air Launch Chaff Countermeasures for the purpose of checking the effect of any design or material change on performance, and to assure adequate quality control. The Air Launch Chaff Countermeasures selected for special inspections may be selected from those previously subjected to sampling inspections.

4.4.3.1 Special inspection schedule. Selection of Air Launch Chaff Countermeasures or assemblies for special inspections shall be made as follows:

a. On early Air Launch Chaff Countermeasures after an engineering or major change.

b. Whenever failure reports or other information indicate that additional inspections are required. (This will be determined by the procuring activity.)

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4.4.3.2 Scope of inspections. Special inspections shall consist of such examinations and tests as specified in the contract (see 4.4.3.2 in 6.2.2). Inspection procedures previously approved for the first article inspection shall be used where applicable.

4.4.4 Air launch chaff countermeasures failure. Should a failure occur during either the sampling or special inspections, the following action shall be taken:

a. Determine the cause of the failure by conducting a failure investigation and analysis.

b. Determine if the failure is an isolated case of design or fabrication defect.

c. Where practical, include an inspection in the individual inspections to check all units (see 6.4.16) for this requirement until assurance is obtained that the defect has been corrected.

4.5 Inspection procedures. The right is reserved by the procuring activity or the Government inspector to modify the inspections or require any additional inspections deemed necessary to determine compliance with the requirements of this specification and the contract (see 6.2.1). MIL-T-18303 shall be used as a guide for preparation of inspection procedures. When approved inspection procedures are available from previous contracts, such procedures may be used provided their use is acceptable to the procuring activity; however, the right is reserved by the procuring activity to require modification of such procedures, including additional inspections, when deemed necessary (see 6.2.1).

4.6 Inspection of material.

4.6.1 Chaff material inspection. Examination of the materials receiving inspection report for each receiving inspection lot (see 6.4.10) and on-line inspection of chaff-cut lots (see 6.4.8) and chaff-hank (see 6.4.5) lots (see 6.4.9) as specified herein shall be made to adequately demonstrate compliance with the requirements of this specification.

4.6.1.1 Chaff-hank inspections.

4.6.1.1.1 Dipole characteristics inspection. Fifty fibers or strands from each chaff-hank lot shall be inspected for the weight, dimension and coating requirements of this specification. If more than 5 fibers or strands do not meet the specified requirements, the lot shall be rejected and production stopped until corrective action has been taken. One chaff-hank from each chaff-hank lot may be weighted, as a unit, in lieu of 50 fibers or strands to verify the in-process weight requirements.

4.6.1.1.2 Secondary (slip) coating. The formulation of the secondary (slip) coating shall be verified by means of appropriate chemical analysis. An incorrect formulation shall be rejected and not used for production until corrective action has been taken.

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4.6.1.1.3 Conductivity and maximum resistivity test of the coating. Randomly select not less than five fibers 36 inches long from a chaff-hank lot. The fibers shall be used to test the conductivity and resistivity of the coating. Using a d.c. ohmmeter, the measurement shall be made on the interior 30 inch length of the 36 inch fiber length for the electrical resistance. Any fiber which fails to meet all the requirements of 3.3.1 shall be cause for rejection of the lot.

4.6.1.2 Chaff-cut/muffin inspection.

4.6.1.2.1 Dipole length inspection. Twenty dipoles from each chaff-cut lot (specified herein) shall be selected at random from each chaff-cut lot. Each of these dipoles shall be measured and recorded. If more than four dipoles from any chaff-cut, or more than 15 percent of the total sample are found to be out of the specified tolerance, that lot of chaff-cuts shall be rejected and production stopped until corrective action has been taken. An alternate method in lieu of dipole length inspection may be used, by measuring the length of the chaff muffin. From each lot of each type of chaff muffin, randomly select five muffins from each type cut. If more than one muffin from any muffin-cut, or more than 15 percent of the total samples are found to be out of specified tolerance, that lot of chaff muffin-cuts shall be rejected and production stopped until corrective action has been taken. The inspection shall be repeated on the first chaff-cut processed after the corrective action has been taken to assure compliance with the specification requirements.

4.6.1.2.2 Chaff muffin dipole inspection. From each lot of each type of chaff muffin, randomly select 5 muffins from each type cut. The chaff muffin shall be inspected after the cutting operation to visually inspect for dipole end-welding, metal smearing and cohesion between dipoles. Any chaff muffin failing to meet all the requirements of 3.3 shall be cause for rejection of the muffin lot.

4.7 Visual inspection.

4.7.1 Chaff payload/assembly(s). Verification of the presence of all component parts in accordance with the applicable DL shall be made through visual inspection.

4.7.2 Air launch chaff countermeasures components. Verification of the presence of all component parts in accordance with 3.8.1 and the applicable DL shall be made through visual inspection.

4.7.3 Air launch chaff countermeasures marking. Visual inspection shall be made to confirm compliance with the applicable specification sheet and DL.

4.7.4 Packaging and packing. Visual inspection shall be made of the packaging and packing to assure compliance with Section 5 of this specification. Any deviations from the specification shall be cause for rejection.

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4.8 Air launch chaff countermeasures inspection.

4.8.1 Air launch chaff countermeasures assembly inspection. Air Launch Chaff Countermeasures shall have an assembly inspection in accordance with the applicable specification sheet (see 3.1).

4.8.2. Weight. All first article, initial production and production chaff payload/assemblies and Air Launch Chaff Countermeasures shall be weighed in accordance with 3.8.4 and recorded.

4.9 Environmental test. The simulated environmental test specified below shall be performed in the order presented herein. Tolerances for these tests shall be in accordance with MIL-STD-810 unless otherwise specified herein. The Air Launch Chaff Countermeasures shall show no signs of degradation or appreciable damage and shall meet all dimensional and performance requirements after exposure.

4.9.1 Temperature and altitude. The Air Launch Chaff Countermeasures shall be subjected to a temperature and altitude test in accordance with MIL-STD-810, Method 520.0, Procedure III using the profile shown on Figure 1. The temperature change rate (vertical Lines on Figure 1) are not meant to be realistic, but shall be dependent on the test facility and shall not effect the one hour dwell time.

4.9.2 Humidity. The Air Launch Chaff Countermeasures shall be subjected to a humidity test in accordance with MIL-STD-810, Method 507.2, Procedure III.

4.9.3 Salt fog. The Air Launch Chaff Countermeasures shall be subjected to a salt fog test in accordance with MIL-STD-810, Method 509.2, Procedure I.

4.9.4 Shock. The Air Launch Chaff Countermeasures shall be subjected to a shock test in accordance with MIL-STD-810, Method 516.3, Procedure I.

4.9.4.1 Shock pulse. The pulse shape shall be as shown on Figure 516.3-1 of MIL-STD-810 except that an Electrodynamic Vibration machine may be used, if so, the tolerances of the specified pulse may be modified to allow amplitude variations up to +/-20 percent of maximum amplitude prior to the initiation of the sawtooth ramp. All other tolerances apply. (The change in tolerance allows for the normal necessary pre- and post-pulses for Electrodynamic Machine Shock Excitation.

4.9.4.2 Shock/vibration mounting. The shock/vibration mounting shall be as specified in the applicable specification sheet (see 3.1). The control accelerometer location for each axis shall be as specified in the applicable specification sheet (see 3.1).

4.9.5 Vibration. The Air Launch Chaff Countermeasures shall be subjected to a vibration test in accordance with MIL-STD-810, Method 514.3, Category 6, Procedure 1, and Figure 2 of this specification except as follows:

a. The Air Launch Chaff Countermeasures must withstand both a Low Level Functional and Endurance Test.

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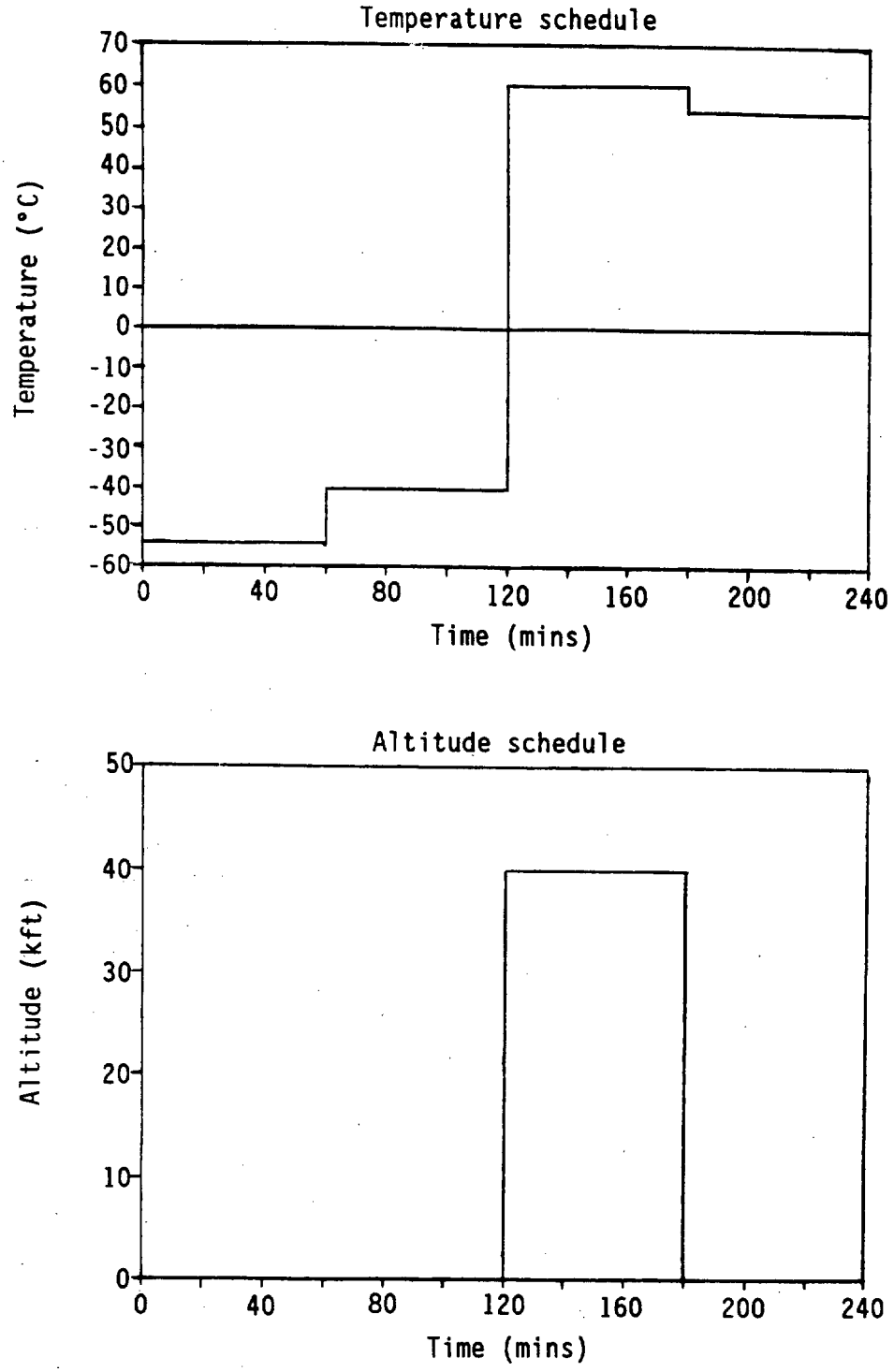


FIGURE 1. Temperature and altitude test profile for the Air Launch Chaff Countermeasures.

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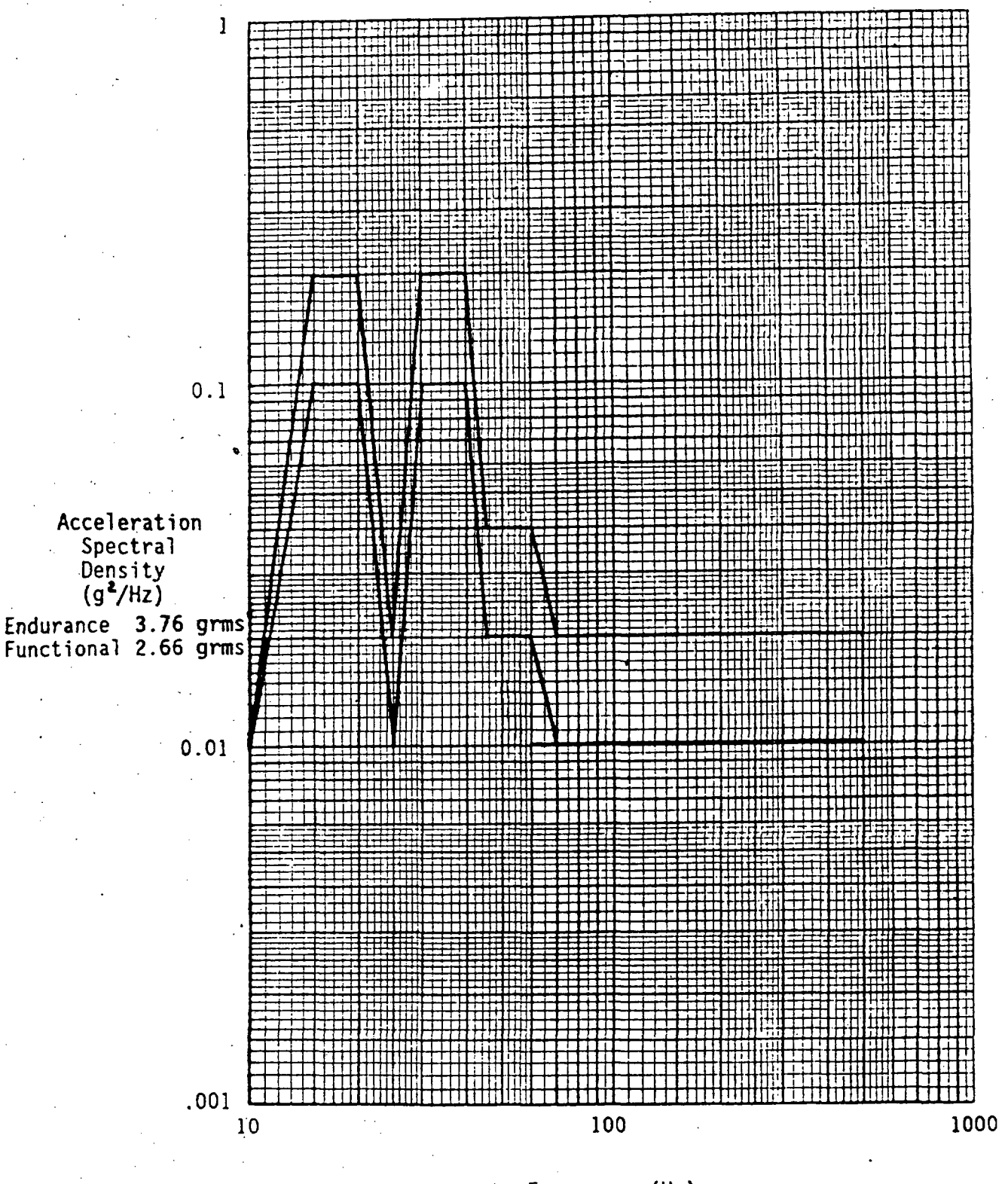


FIGURE 2. Qualification vibration spectrum for the Air Launch Chaff Countermeasures.

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b. Testing shall be performed in all three axis.

c. The same mounting procedure used for shock testing shall be used for the vibration test.

4.9.5.1 Vibration test sequence.

a. Functional Vibration as shown Figure 2 for 30 minutes in each axis.

b. Endurance Vibration as shown on Figure 2 for one hour in each axis.

c. Repeat Step a.

4.10 Functional test.

4.10.1. Flight test. The Government performed flight test will be conducted on first article, initial production, quality conformance, and special test units to verify performance capability. Tests will include Radar Cross-Section (RCS) (see 6.4.14) measurements and "A" size Sonobuoy launch platform/dispenser compatibility tests in accordance with MIL-STD-2071, this specification and the applicable specification sheet (see 3.1).

4.10.1.1 Aircraft. Flight test will be conducted by the Government using an aircraft type which has previously been used to provide data to establish the minimum performance requirements specified in MIL-STD-2071 and this specification.

4.10.1.2 Launcher/dispenser. The Air Launch Chaff Countermeasures RR-( )/AL will be launched from the "A" size Sonobuoy launch platform/dispenser in accordance with its respective technical manual and this specification.

4.10.1.3 Loading launcher. The Air Launch Chaff Countermeasures RR-( )/AL, with or without a Chaff Launch Container, depending on platform, will be loaded in the "A" size Sonobuoy launch platform/dispenser in accordance with its respective technical manual.

4.10.1.4 Radar. The Air Launch Chaff Countermeasures RR-( )/AL will be evaluated using radar equipment in accordance with this specification.

4.10.1.5 Sampling. The test objective will be to collect RCS data in all 100 gates for all test samples. One Air Launch Chaff Countermeasures shall be launched per flight test run in accordance with this specification.

4.10.1.6 Data processing of pulse to pulse RCS-data. The pulse-to-pulse RCS data will be processed as specified in MIL-STD-2071 to provide: shadow graphs, first level statistics, including summary sheets; and second level statistics for basic data, corrections for aspect angle and corrections for aspect angle and antenna gain. A printout shall be provided to show range, azimuth angle, elevation angle and aspect angle for all flight tests.

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4.10.1.7 Flight test failure criteria. Any Air Launch Chaff Countermeasures that fails to function properly in accordance with the requirements of 4.10.1 and this specification shall be cause for rejection of the lot.

4.10.2 Radar Cross-Section procedures.

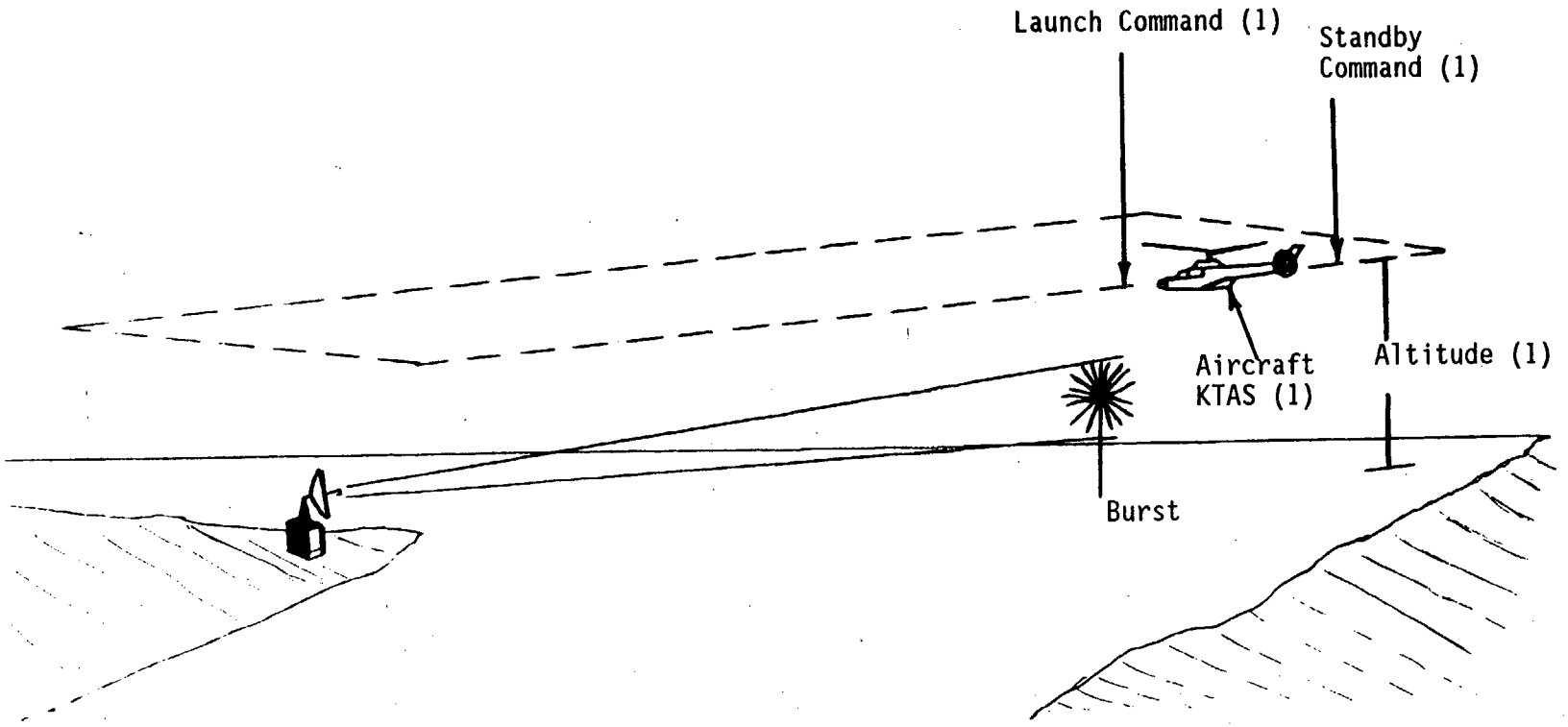
4.10.2.1 Aircraft operation. The aircraft shall fly a "racetrack" course as depicted on Figure 3, a standby and launch command will be given at a range specified in the specification. The aircraft will follow a course directly over the test radar. The Air Launch Chaff Countermeasures will be launched on the inbound radial on command. Upon initial radar acquisition, the aircraft shall be directed to maximum range, then brought in for each run.

4.10.2.2 Data gathering. The test radar will track the aircraft during the inbound radial run until the launch command is given. At that time the test radar will position its beam to the range, azimuth and elevation where the aircraft was when the launch command was given. Data gathered where the combination of aspect angle and range makes correction of RCS values questionable will be identified and rejected as invalid data. The range in yards divided by the sine of the aspect angle will be equal to or greater than specified in the applicable specification sheet for acceptable data in gates 1 thru 100. As depicted on Figure 3, a standby and launch command will be given at a range specified in the applicable specification sheet. Data recording will commence at the launch command and will continue for 60 seconds after the launch command. If old chaff creates an interference problem, a delayed start will be acceptable. RCS data will be collected from gates 1 thru 100 which will be spaced by the number of nanoseconds in time specified in the applicable specification sheet (see 3.1).

4.10.2.3 Data evaluation. Data shall be evaluated to determine the presence of not less than 100 gates of pulse-to-pulse data for each chaff unit meeting the requirements of the procedures for collecting data on old chaff, except there will be no spacing between gates, and the requirements for aspect angle as specified in MIL-STD-2071. The shadow graph shall be examined to ensure that there is a minimum of ninety gates of data for each unit to meet the requirements for valid data as stated in 4.10.2.2. Poor radar tracking in angle as shown by the shadow graph shall invalidate the data that would otherwise meet the requirements of 4.10.2.2. The valid data as shown by the shadow graphs shall be identified for further data processing. The test data shall be invalid for any chaff unit where less than 90 gates contain valid RCS data.

4.10.2.4 Data processing. The RCS data shall be processed by searching the 100 received gates,and selecting the peak RCS value from these 100 gates for each return pulse. The selected peak RCS value for each pulse shall be averaged for 240 radar pulses to arrive at a one-half second "average peak" RCS. These one half second "average peak" RCS values shall be plotted as a function of time. The plots shall depict the radar data gathering time from launch command till 60 seconds after launch command. Processing shall be accomplished in accordance with MIL-STD-2071 and provide the outputs specified in 4.10.1.6 herein. The data shall be processed to provide average RCS values and the weighted average of the number of consecutive gates specified in 4.10.2.2

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(1) Refer to applicable specification sheet for flight profile data.

FIGURE 3. Flight profile for testing the Air Launch Chaff Countermeasures.

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4.10.2.4.1 Calibration check. RCS measurement of a standard calibration sphere shall be made prior to any chaff measurements. The sphere RCS data shall be compared to the theoretical RCS of the sphere and the calibration constant(delta K) for processor shall be set to provide the measured sphere data as near as possible to the theoretical value of the sphere RCS. The corrected value of delta K shall be used in the first and second level data processing.

4.10.3 Pass/fail criteria. The Air Launch Chaff Countermeasures RR-( )/AL shall meet the requirements of this specification and the applicable specification sheet (see 3.1).

4.10.3.1 Radar cross-section. The average of the pulse-to-pulse RCS values as measured shall meet or exceed the number of square meters specified in this specification for any unit to be acceptable.

5. PACKAGING

5.1 General. All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or purchase order (see 6.2.1) in accordance with MIL-STD-794 and as specified herein. Unless otherwise specified in the contract, the method of preservation-packaging shall be Method I, Submethod IA-15 in accordance with MIL-P-116 and this specification (see 6.2.1).

5.2 Packaging details.

5.2.1 Chaff launch container. The Air Launch Chaff Countermeasures shall be installed in a Chaff Launch Container in accordance with the applicable specification sheet (see 3.1).

5.2.2 Shipping container. The Air Launch Chaff Countermeasures shall be packaged in accordance with the requirements of this specification, applicable specification sheet (see 3.1) and applicable DL.

5.2.3 Marking. The Air Launch Chaff Countermeasures shall be marked in accordance with MIL-STD-129, CFR 49 parts 171-173 and applicable DL.

5.2.4 Marking for shipment. The shipping container shall be marked in accordance with the applicable specification sheet (see 3.1), applicable DL and as specified in the contract (see 6.2.1).

5.3 Palletization. Palletizing shall be accomplished in accordance with MIL-STD-1323, applicable specification sheet (see 3.1), DL and contract (see 6.2.1).

6. NOTES

6.1 Intended use. The Air Launch Chaff Countermeasures RR-( )/AL is a chaff loaded unit and is intended to be used to dispense a chaff cloud to create a simulated target, decoy, or other types of radar confusion. The Air Launch Chaff Countermeasures is intended for air launch from "A" size Sonobuoy Launch Platforms in various aircraft.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Air Launch Chaff Countermeasures type (see item description of applicable specification sheet).

c. Levels of preservation-packaging, packing and marking required (see Section 5).

d. If design, construction or workmanship are to be other than specified in the specification (see 3.4).

e. If first article is required (see 3.2, 6.3).

f. If test procedures are to be expanded or modified (see 4.5).

6.2.2 Data requirements. When this specification is used for an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph No.	Data Requirement Title	Applicable Option DID No.
3.2, 4.5	Procedure, First Article Inspection	DI-T-4901
4.2	Report, First Article Inspection	DI-T-4902
4.4	Report, Production Inspection	DI-T-4904
4.4.3.2, 4.5	Procedure, Production/Acceptance Inspection	DI-T-4903
4.4.4	Failure Data Collection, Analysis and Corrective Action Plan	UDI-T-23719

(Data item description related to this specification and identified in Section 6 will be approved and listed as such in DOD 5010.12L, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.3 First article. When a first article inspection is required, the items shall be a first article sample. The first article should consist of 12 Air Launch Chaff Countermeasures as required per applicable specification sheet (see 3.1). The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results and disposition of first article. Invitations for bids should provide that the Government reserves the right to waiver the requirement for samples for the first article inspection to those bidders offering a product which has been previously acquired or tested by the Government and that bidders offering such products, who wish to rely on such production to test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions. The following definitions apply to this specification:

6.4.1 Air launch chaff countermeasures. The configuration consisting of the RR-( )/AL.

6.4.2 Chaff launch container. An interface device that allows the Air Launch Chaff Countermeasures to be launched from "A" size Sonobuoy launch platforms.

6.4.3 Chaff payload. Contains a specified quantity of chaff muffins comprised of small diameter, aluminum-coated, glass fibers of specific length requirements wrapped in a specified film.

6.4.4 Chaff-cut. A process by which a quantity of dipoles cut from a chaff-hank to a specific length to provide dipole elements for placement within an assembly, normally a muffin.

6.4.5 Chaff-hank. An assemblage of drawn aluminized glass fibers produced by one machine prior to the dipole cutting operation. The number of fibers in the chaff-hank is normally equivalent to the number of dipoles required for each chaff-cut within a specific chaff muffin.

6.4.6 Dipole (chaff). Small diameter, aluminum-coated, glass fibers, cut to specified radar threat frequencies.

6.4.7 Lot, air launch chaff countermeasures. A Air Launch Chaff Countermeasures lot shall contain that quantity of Air Launch Chaff Countermeasures as specified in the contract or purchase order.

6.4.8 Lot, chaff-cut. A chaff-cut lot shall consist of all chaff-cuts produced by one or more production machines for use in the Air Launch Chaff Countermeasures payload. The lot shall contain that quantity of chaff-cuts necessary to produce one Air Launch Chaff Countermeasures lot.

6.4.9 Lot, chaff-hank. A chaff-hank lot shall consist of all chaff-hanks produced by one or more production machines within the same time frame for use at one time. The lot shall not contain more hanks than that amount required to produce one lot of Air Launch Chaff Countermeasures.

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6.4.10 Lot, receiving inspection. A receiving inspection lot shall consist of all material of one type received from one supplier at one time.

6.4.11 Metal smear. Cohesion of adjoining metal surfaces, as a result of the manufacturing process.

6.4.12 Muffin. A quantity of small diameter dipoles (chaff-cut aluminum-coated glass fibers) bundled in a polyimide wrap. In some cases two or more of these bundles are arranged end-to-end and are provided with an additional polyimide wrap to form a muffin pack. For the purpose of this specification the muffin is used to describe an independent bundle of dipoles or each bundle of dipoles within a muffin pack.

6.4.13 Muffin pack. Two or more muffins arranged end-to-end and wrapped in a polyimide film.

6.4.14 Radar Cross-Section (RCS). An average value of the individual pulse measurement of the chaff cloud area.

6.4.15 Slip coating. A secondary coating, in layer form, which promotes non-cohesion, i.e.; dipole fibers are prevented from sticking together.

6.4.16 Units. As used herein refers to a Air Launch Chaff Countermeasures or a group of Air Launch Chaff Countermeasures.

6.5 Dipole quantity per chaff-cut. The dipole quantity per chaff-cut, (Q) can be calculated, given the dipole diameter (d) and the percent packing density (p), as follows: $Q = \frac{p}{100} \cdot \frac{1.44414}{2.59808\ (0.57735d)}$

6.6 Test equipment. NONE

6.7 Safety.

6.7.1 Explosive safety precautions. Minimum explosive safety precautions for use by the contractor are detailed in DOD Instruction 4145.26M DOD Contractors Safety Manual for Ammunition, Explosives and Related Dangerous Materials. In addition, it is the contractors responsibility to comply with all local, state and Federal Regulations concerning personnel health and safety regarding the use of hazardous materials including but not limited to explosive and pyrotechnic materials.

6.8 Subject term (key word) listing.

1. Chaff Countermeasures, Air Launch
2. Coating, slip
3. Container, Chaff Launch
4. Cut, chaff

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5. Dipole
6. Hank, chaff
7. Lot, chaff-cut
8. Lot, chaff-hank
9. Lot, Air Launch Chaff Countermeasures
10. Metal smear
11. Muffin
12. Muffin pack
13. Payload, chaff
14. Radar Cross-Section
15. Unit

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-85699(AS)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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